SS2P2: A Stable State-Space Point Process for Limit-Order-Book Simulation

Anonymous submission

Abstract

Simulating limit-order-book (LOB) order flow is a fundamental generative task for market-making world models, agent training, and counterfactual market analysis. Neural marked temporal point processes (MTPPs) predict the next LOB event accurately, but their simulation behaviour is largely unexamined: closed-loop, the learned self-excitation drifts or explodes, and we show that the standard remedy of rescaling the intensity to match the real event rate often has *no solution* for unbounded models. To close this gap we propose SS2P2 (Stable State-Space Point Process), which keeps the expressive state-space backbone of S2P2 and replaces only the output heads: a *softmin-bounded rate head* whose intensity carries a hard closed-form ceiling (an exact dominating rate for Ogata thinning) with a floor of exactly zero, and a *rate-neutral* soft-max mark head that can be arbitrarily deep without perturbing the total rate. Extensive experiments on three Coinbase assets, against five baselines with three training seeds each, lossless streaming evaluation, and *verified* rate calibration for every model, show that SS2P2 is the strongest predictor on two of three assets while remaining verifiably calibratable closed-loop with competitive simulated structure. Our sharpest finding is that whether a model *can* be calibrated is itself a measurable property. The unbounded parent model fails verified calibration on 7 of 9 checkpoints (on one asset, no checkpoint can be calibrated at all) because its simulated rate responds almost discontinuously to the rescaling constant; the bounded head passes on all nine. A per-gap decomposition shows that the remaining prediction gap comes almost entirely from quiet periods, which is the expected and accepted cost of bounding the intensity. SS2P2 trains with an exact parallel scan, simulates with $O(1)$ cost per event using Ogata thinning, and matches real event rates after verification.

Introduction

The limit order book (LOB) is the core mechanism of electronic markets: a continuous-time stream of order placements, cancellations, and trades whose fine structure drives price discovery and liquidity. Two tasks are asked of a generative model of this stream. The first is *prediction*: given the event history, score the time and type of the next event. The second is *simulation*: roll the model forward freely, feeding its own samples back as history, and reproduce the stylized facts of real order flow, such as calibrated event rates, burst–lull clustering, and heavy-tailed returns (Cont 2001). Simulation is the harder task and, for downstream use (market-making world models, agent training, counterfactual analysis), the more valuable one (Li et al. 2025; Nagy et al. 2025).

Neural MTPPs (Du et al. 2016; Mei and Eisner 2017; Shchur et al. 2021) excel at prediction, and continuous-time state-space variants (Chang et al. 2025; Shi and Cartlidge 2022) push held-out likelihood to the state of the art. But that literature evaluates almost exclusively teacher-forced. The S2P2 paper itself (Chang et al. 2025) reports state-of-the-art held-out likelihood yet never rolls the model out, leaving closed-loop simulation an open gap. Two obstacles stand between a trained MTPP and a usable simulator. *First, instability*: the self-excitation that wins the one-step likelihood becomes a liability closed-loop, because each sampled event raises the intensity, which produces more events, and with no mechanism bounding the rate the roll-out drifts hot or explodes. *Second, uncalibratability*: the standard patch, rescaling the intensity until the simulated rate matches reality, turns out to be unavailable exactly where it is needed most. We find trained unbounded state-space checkpoints whose closed-loop rate is *near-discontinuous* in the rescaling constant (a 10^{-3} change flips the rate across the entire tolerance band), so that no constant rescaling exists; on one of our three assets, this leaves the unbounded model with no usable simulator at all. The field thus faces a trade-off between expressivity and stability: expressive models predict well but cannot be trusted (or even calibrated) in closed loop, while simple stable models simulate safely but predict poorly.

SS2P2 changes only the output heads of S2P2; the backbone is untouched. We keep the stacked latent-linear-Hawkes state-space backbone verbatim, with its diagonal zero-order-hold dynamics, parallel-scan computability, and LayerNorm'd stack output h_t, and replace only what comes after h_t: a **softmin-bounded rate head**, whose intensity has a hard closed-form ceiling (an exact dominating rate for Ogata thinning) and an exactly-zero floor with unit log-intensity gradient in quiet periods; and a **rate-neutral mark head**, a soft-max over event types that cannot change the total rate no matter how deep its network is.

Because the backbone is identical, differences against the unbounded parent (S2P2) are attributable to the heads. Our contributions:

1. **The SS2P2 head pair** (softmin-bounded rate $\times$ rate-

neutral marks). It turns S2P2, which is state-of-the-art teacher-forced but unstable and often uncalibratable in free-running simulation, into a *provably stable simulator*: a bounded, thinning-exact intensity on the unmodified state-space backbone, trained with stateful (TBPTT) likelihood and simulated with $O(1)$-per-event carried state.

2. **Calibratability as a measurable model property.** Every model is rate-calibrated by a mark-preserving bisection with *full-scale verification*: the unbounded head fails on 7/9 checkpoints with a critical-point signature, including all three seeds on SOL where no usable simulator remains, while the bounded head verifies on all nine. To our knowledge this is the first cross-asset quantification of closed-loop calibratability for neural TPPs.
3. **Exact post-hoc rate calibration** for factorized heads. The rate is linear in a learnable scale, the ceiling scales with it, and the mark distribution is untouched, so calibration is a one-dimensional root-find that keeps the guarantee intact and *improves* every structure metric by correcting the simulated clock rate.

Background and Related Work

Marked temporal point processes. An MTPP models a sequence $\mathcal{H}_T = \{(t_i, e_i)\}_{i=1}^{N_T}$ of events with times t_i and types $e_i \in \mathcal{E}$, a finite set of atomic event types, through per-type conditional intensities $\lambda_e(t \mid \mathcal{H}_t)$, conditioned on the history $\mathcal{H}_t$ of all events up to time t (model parameters θ suppressed; we abbreviate $\lambda_e(t)$ when the conditioning is clear). Writing $\lambda(t) = \sum_{e \in \mathcal{E}} \lambda_e(t)$ for the total (ground) intensity and $p(e \mid t) = \lambda_e(t)/\lambda(t)$ for the mark distribution, the log-likelihood of a sequence is

$$\log \mathcal{L} = \sum_i \Big[\log \lambda(t_i) + \log p(e_i \mid t_i)\Big] - \int_0^T \lambda(u \mid \mathcal{H}_u)\, du,$$

the integral being the *compensator* $\Lambda(t) = \int_0^t \lambda(u \mid \mathcal{H}_u)\, du$. By the random time change theorem, the rescaled masses $u_i = \Lambda(t_i)$, $\Delta u_i = u_i - u_{i-1}$ are i.i.d. $\mathrm{Exp}(1)$ iff the intensity is correctly specified; we report their mean (target 1) and KS distance as timing-calibration diagnostics. Sampling uses Ogata thinning, which requires a dominating rate $\bar{\lambda} \geq \lambda(t)$; a model whose intensity has no computable upper bound must resort to heuristic (and bias-prone) dominating-rate estimates.

The S2P2 backbone. S2P2 (Chang et al. 2025; Shi and Cartlidge 2022) is a state-space point process: L stacked latent-linear-Hawkes/SSM layers with diagonal state, ZOH-discretized between events and computable by parallel scan,

$$x_{t'}^{(\ell)} = \bar{A}^{(\ell)} x_t^{(\ell)} + (\bar{A}^{(\ell)} – I)\, \tilde{B}^{(\ell)} h^{(\ell-1)}\ \big[+\, \tilde{E}^{(\ell)} \alpha \text{ at an event}\big],$$

with a LayerNorm'd residual readout $h^{(\ell)} = \mathrm{LayerNorm}\big(\mathrm{GELU}(y^{(\ell)}) + h^{(\ell-1)}\big)$ producing the continuous-time hidden representation $h_t := h^{(L)}(t) \in \mathbb{R}^H$. The published model reads the rate from h_t through a projection-and-softplus head, $\lambda = \mathrm{softplus}(w^\top h_t + b)$, which is unbounded above and has no learnable scale; marks are coupled to the same rate field (our benchmark implementation of this baseline splits h_t into disjoint halves for the rate and mark heads). SS2P2 keeps everything up to and including h_t and changes only the two heads.

Evaluation axes. We score *prediction* per genuine test event (NLL splits, type accuracy, expected-time MAE, time-rescaling diagnostics) and *simulation* by calibrated closed-loop roll-outs scored on stylized facts of real order flow (Cont 2001; Vyetrenko et al. 2020); the protocol is detailed in the Experiments section.

Neural TPPs for event streams. Neural MTPPs learn history-dependent intensities with recurrent (Du et al. 2016; Mei and Eisner 2017), attention-based (Zhang et al. 2020; Zuo et al. 2020), jump-diffusion (Zhang et al. 2024), and continuous-time state-space (Chang et al. 2025) encoders (survey: Shchur et al. 2021). Shi and Cartlidge (2022) specialize neural Hawkes models to LOB event streams and evaluate both prediction and simulation. Notably, the state-space line is evaluated almost exclusively teacher-forced: Chang et al. (2025) report state-of-the-art held-out likelihood but never roll the model out (they avoid thinning altogether). Closed-loop simulation is an unaddressed gap of the S2P2 paper, and our experiments show it is not benign: the unbounded head is frequently *uncalibratable* in free roll-out (Table 3). Our contribution is orthogonal: we keep the backbone verbatim and change only what is read *out* of it.

Hawkes modeling of order flow. Parametric Hawkes processes are the classical LOB event model (Hawkes 1971, 2018; Jain et al. 2024a,b), with state-dependent extensions (Morariu-Patrichi and Pakkanen 2022). Their linear structure gives exact stationarity certificates but limited expressiveness; SS2P2 takes the complementary route of full neural expressiveness with stability restored by a closed-form bound. In the neural TPP literature such bounds appear mainly as computational devices for thinning; here the bound is a modeling decision with an asymmetric requirement (hard ceiling, exactly-zero floor), validated by our calibration study.

Generative LOB simulators. Learned order-flow generators span GANs (Li et al. 2020; Coletta et al. 2021, 2022), token-level autoregressive state-space models (Nagy et al. 2023), diffusion-guided agents (Huang et al. 2026), and foundation-model-scale transformers (Li et al. 2025; Sood et al. 2026); agent-based simulators (Byrd, Hybinette, and Balch 2020; Frey et al. 2023) remain the RL workhorse. These systems are scored on distributional realism: stylized facts (Cont 2001; Vyetrenko et al. 2020) and the KS/Wasserstein suites of LOB-Bench (Nagy et al. 2025), whose authors report error accumulation over long roll-outs. SS2P2 differs on two counts: one likelihood-based model is held to *both* prediction and closed-loop realism, and the simulated rate is calibrated and *verified*, a closed-loop stability certificate that no generative simulator above carries.

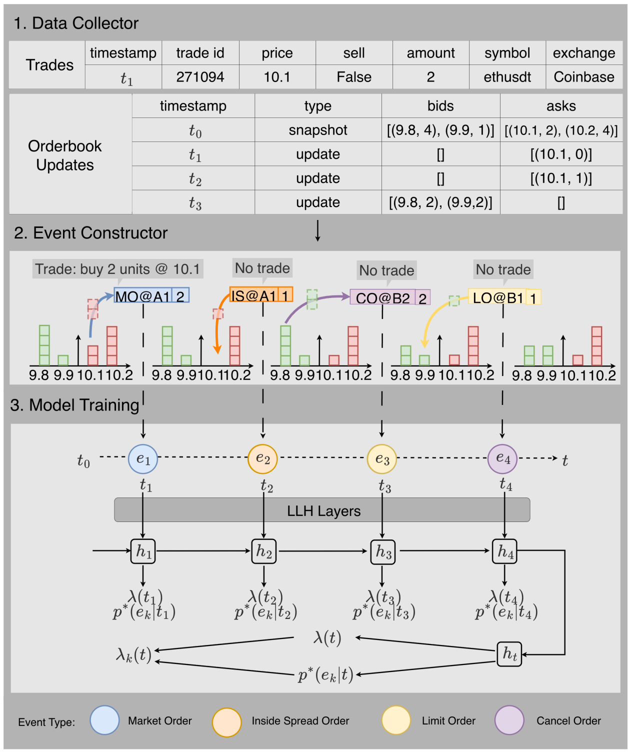

Figure 1: Pipeline. (1) *Data collector*: synchronized trade records and Level-2 order book updates. (2) *Event constructor*: consecutive book snapshots are differenced and aligned with trades to emit atomic typed events. (3) *Model training*: the event sequence drives the state-space backbone (LLH layers) whose continuous-time hidden representation h_t decodes the bounded total rate $\lambda(t)$ and the rate-neutral mark distribution $p(e \mid t)$, giving per-type intensities $\lambda_e(t) = \lambda(t)\, p(e \mid t)$.

Methods

Event constructor

To apply point process modeling to high-frequency limit order book (LOB) data, we represent market dynamics as a sequence of **events** (pipeline overview in Figure 1). We consider the top $K = 10$ levels on each side (bid and ask).

Atomic Events. We define a finite set $\mathcal{E}$ of *atomic event types*, each denoted $e = (c, s, \delta)$, where:

- $c \in \{\mathrm{LO}, \mathrm{MO}, \mathrm{CO}, \mathrm{IS}\}$ is the *event class*:
  - **Limit Order (LO)**: liquidity provision at one of the top-K levels.
  - **Cancel Order (CO)**: liquidity withdrawal at one of the top-K levels.
  - **Market Order (MO)**: liquidity consumption at the best quote.
  - **Inside-Spread Order (IS)**: a new order placed inside the spread, improving the prevailing best quote.
- $s \in \{b, a\}$ is the *side* (bid or ask).
- δ is the *level index*:

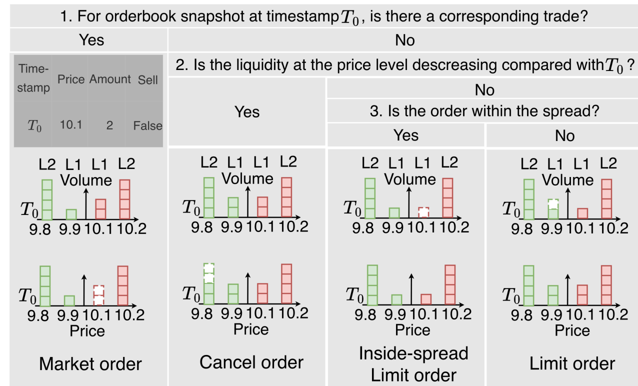

Figure 2: Event construction for the four order classes through three questions, each a decision-tree classification applied to synchronized Level-2 book snapshots and trade records at timestamps $T_0 \to T_1$.

  - For LO/CO, $\delta \in \{1, \ldots, K\}$ indicates distance from the best quote ($\delta = 1$ = best bid/ask, $\delta = 10$ = tenth level).
  - For MO, $\delta = 1$ always, since market orders execute against the current best quote.
  - For IS, $\delta \in \{1, \ldots, K\}$ denotes the number of ticks by which the new quote improves the spread (e.g., $\mathrm{IS}_{b,2}$ = new bid order two ticks inside the spread).

Event Construction from LOB and Trade Data. As illustrated in Figure 2, each atomic order event is inferred by differencing consecutive Level-2 order book snapshots and aligning them with trade records at identical timestamps. Because trades and book updates are perfectly synchronized in both time and volume, every observed change can be classified without ambiguity through a three-step decision tree. First, if the best-quote liquidity decreases and a trade is recorded at the same price, the change corresponds to a **Market Order**, a direct execution against standing quotes. Second, if liquidity decreases at any level without a matching trade, the reduction represents a **Cancel Order**, reflecting liquidity withdrawal. Finally, when liquidity increases, the classification depends on the price position: if the new order appears inside the spread, it forms an **Inside-Spread Order**; otherwise, it is a **Limit Order** placed within the existing book depth. This rule set ensures exhaustive mapping from raw order book and trade data into atomic order events, enabling a consistent foundation for downstream temporal modeling.

Stable State-Space Point Process (SS2P2)

SS2P2 factors the per-type intensity into a bounded total rate and a rate-neutral mark distribution, the ground-intensity–mark factorization of set-valued MTPPs (Chang, Boyd, and Smyth 2024), both decoded from the *same* continuous-time hidden representation h_t:

$$\lambda_e(t) \;=\; \lambda(t) \cdot p(e \mid t), \qquad \sum\nolimits_{e \in \mathcal{E}} \lambda_e(t) = \lambda(t),$$

the second identity holding because $\sum_e p(e \mid t) = 1$. The design principle is to put nonlinearity where it cannot perturb the rate.

Softmin-bounded rate head

For s2p2, the closed-loop simulation fails through a feedback loop: each sampled event raises the intensity, a higher intensity produces events faster, and with an unbounded head the roll-out drifts hot or explodes. The rate head is built to break this loop. From the backbone's continuous-time hidden representation h_t (LayerNorm'd, hence bounded in scale), a gate produces a bounded state g, which is read out through a smooth one-sided cap with fixed constant $z_{\max}$:

$$\begin{aligned}
o &= \sigma(W_o h_t + b_o) \in (0,1)^H, \\
g &= o \odot \tanh(h_t) \in (-1,1)^H, \\
z_{\text{raw}} &= w^\top g + b \ \ \text{(unconstrained)}, \\
z &= z_{\max} - \operatorname{softplus}\big(z_{\max} - z_{\text{raw}}\big), \\
\lambda(t) &= \varsigma \cdot \operatorname{softplus}(z)
\end{aligned}$$

with a learnable positive scale ς initialized so the baseline rate matches a target rate ($g \to 0 \Rightarrow \lambda \approx \varsigma \ln 2$). Above makes four deliberate design choices:

- **Bounded input.** The readout sees $g \in (-1,1)^H$, not the raw latent state: however far the closed loop drives h_t, including states training never visited, an extreme state can change the direction of g but not its magnitude.
- **Hard ceiling.** Since $\operatorname{softplus}(x) > 0$ for every x, $z \le z_{\max}$ holds identically, so $\lambda(t) \le \varsigma \cdot \operatorname{softplus}(z_{\max})$ for every state and every parameter setting; no training outcome can remove the bound. The worst the roll-out can ever do is behave like a homogeneous Poisson process at the ceiling, and the ceiling doubles as an *exact* dominating rate for Ogata thinning.
- **Vanishing gain.** Differentiating the cap gives $\partial z/\partial z_{\text{raw}} = \sigma(z_{\max} - z_{\text{raw}})$: as a burst pushes the intensity toward the ceiling, the head becomes exponentially *less* sensitive to further state excursions, so the feedback loop loses gain exactly when it is most dangerous. This is what makes the closed-loop rate respond smoothly to rescaling, and hence what makes the model calibratable.
- **Open floor.** The cap is one-sided: as $z_{\text{raw}} \to -\infty$, $z \to z_{\text{raw}}$, so $\lambda \sim \varsigma\, e^{z_{\text{raw}}} \to 0$ and $\log \lambda$ keeps unit gradient however quiet the market gets.

The first three choices own simulation stability and calibratability; the fourth ensures the bound costs nothing in prediction when the market is quiet.

Rate-neutral mark head

Marks are decoded from the same h_t but never change the total rate:

$$p(e \mid t) = \operatorname{softmax}\big(\operatorname{MLP}(h_t)\big)_e, \qquad \lambda_e(t) = \lambda(t)\, p(e \mid t).$$

Because the soft-max lives on the simplex, the mark network can be arbitrarily deep (or later conditioned on exogenous state such as an agent's resting quotes) without touching λ. This decouples the two failure modes: rate stability is owned entirely by the bounded head, mark expressiveness entirely by the simplex.

Likelihood and sampling

Training minimizes the negative log-likelihood of each training window, covering its events (t_i, e_i) and the next event after the window:

$$\mathcal{L} = \int_0^T \lambda(u \mid \mathcal{H}_u)\, du \; - \; \sum_i \Big[\log \lambda(t_i) + \log p(e_i \mid t_i)\Big],$$

with T the next-event time. The integral and the $\log \lambda$ terms score *when* events happen (the compensator charges for intensity placed where no event occurred); the $\log p$ term is a categorical cross-entropy scoring *what* each event is. By the factorization, the timing terms train only the bounded rate head and the mark term trains only the soft-max head. Each $\lambda(t_i)$ is evaluated at the left limit, the state just before event i, so an event never informs its own intensity. The compensator integral is approximated by one intensity evaluation per inter-event interval (the endpoint rule); this under-weights a decaying intensity and biases the learned rate upward, which the post-hoc calibration below corrects. Closed-loop sampling uses Ogata thinning with the exact dominating rate $\varsigma \cdot \operatorname{softplus}(z_{\max})$: propose arrivals from a homogeneous Poisson process at the ceiling and accept each with probability $\lambda(t)/\big(\varsigma \operatorname{softplus}(z_{\max})\big)$.

Carried-state simulation

The standard roll-out protocol re-encodes a sliding window of the last W events from a zero state at every step. This is faithful to windowed training but has two costs: $O(W)$ work per simulated event, and a hard memory ceiling. Nothing older than W events can influence the future, so long-memory stylized facts (the slow decay of $|r|$ autocorrelation) are *architecturally* unreachable regardless of the model. Because the backbone is a state-space recursion, both costs vanish: we carry the packed decoder state and advance it with only the new event, which is mathematically identical to encoding the entire history from the seed, at $O(1)$ cost per event: an order of magnitude faster than window re-encoding, with no memory ceiling. Attention- and window-shaped baselines have no such Markov state and retain the sliding window.

Stateful (TBPTT) training

Windowed maximum likelihood encodes every training window from a cold initial state, so the model only ever trains in a "history just began" transient and MLE explains the observed rate with an inflated baseline; the free roll-out, whose state is warm, then double-counts. We train instead with truncated backpropagation through time: batch lanes walk contiguous stretches of the stream in order, and the decoder state at the end of window t is passed, *detached*, as the initial state of window $t{+}1$ (resets only at genuine stream boundaries, to the learned initial state). Gradients remain window-local; state values flow from the start of the stream. This makes the training regime consistent with carried-state simulation, at unchanged per-step cost; it requires non-overlapping windows (stride = window length), so our controls match that setting.

Post-hoc rate calibration

The factorization buys one more property: $\lambda = \varsigma \cdot \mathrm{softplus}(z)$ is *exactly linear* in the learnable scale ς, the thinning ceiling $\varsigma \cdot \mathrm{softplus}(z_{\max})$ scales with it, and the mark head is untouched by construction. The free-running rate, which is monotone but nonlinear in ς through the closed-loop feedback, can therefore be calibrated by a one-dimensional geometric bisection on a multiplier κ: roll out briefly at $\kappa\varsigma$, compare the realized rate to the target, and iterate over a handful of probe roll-outs. The result is a certificate-preserving, sampling-exact simulator at any prescribed mean rate. For *any* decoder, a constant κ multiplying the sim-time intensity is still mark-preserving (type probabilities λ_e/λ are unchanged), so the same bisection applies to every baseline; what is lost off-family is only the exactness of the thinning bound. We make the procedure falsifiable: the target is the *validation*-split rate, the bisection must bracket and converge, and a full-scale validation roll-out must verify the target within tolerance (failures escalate to full-scale probes; unresolved failures exclude the checkpoint and are reported). Transfer to the test split is reported, never gated. We use calibration as a *post-hoc* correction for the residual rate inflation caused by the biased endpoint-rule training compensator, and, in the experiments, as a measurement instrument in its own right.

Experiments and Evaluations

In this section, we describe the experimental setup and evaluate SS2P2 against five baselines on three questions: (1) does the bounded head cost predictive accuracy? (2) does SS2P2 reproduce the structure of real order flow in closed-loop simulation? (3) can each model be rate-calibrated at all? We close with an analysis of computational efficiency.

Experimental Setup

Datasets. We benchmark on Coinbase BTC, ETH, and SOL event-driven order flow, seven days per asset, encoded by the event constructor into $|\mathcal{E}| = 62$ atomic types ($20 + 20 + 2 + 20$ over LO/CO/MO/IS, sides, and levels). Validation-split rates are 38.3, 47.9, and 24.0 events/s: production event rates, not subsampled streams.

Baselines. All models share the same input embedding and training configuration; they differ only in the decoder.

- **NHP** (Mei and Eisner 2017): the neural Hawkes continuous-time LSTM.
- **LSTM**: a standard LSTM decoder (adapted generic backbone, not a paper-faithful re-implementation).
- **SAHP** (Zhang et al. 2020): a causal-attention decoder (adapted generic backbone).
- **PCT-LSTM**: a per-type parallel continuous-time LSTM.
- **S2P2** (Chang et al. 2025): the unbounded state-space parent; identical backbone to SS2P2, unbounded softplus head.

Training and evaluation protocol. Each model trains per asset with sequence length 1024, non-overlapping windows, identical gradient-step counts, and three training seeds. The fixed 1024-event context spans only $\sim$20–40 s of market time at these rates, so long-memory structure leans entirely on carried state. Evaluation is strict end-to-end:

- **Prediction** is scored on *every* genuine test event by a lossless streaming evaluator with carried state (one cold start per genuine segment boundary); the evaluator hard-fails if any event goes unscored.
- **Simulation** uses carried-state closed-loop roll-outs (32×600 s, three roll-out seeds per checkpoint) against equal-duration bootstrap samples of real segments. SS2P2 samples by Ogata thinning with its exact closed-form ceiling; baselines, which admit no exact bound, share a common compensator-inversion sampler.
- **Calibration**: every model is rate-calibrated before scoring by a one-dimensional bisection on the time-rescaling constant κ (mark-preserving for every decoder; see Methods), targeting the *validation*-split rate; a full-scale validation roll-out must then verify the target within 15%. Checkpoints that fail are excluded and the failure is reported as a result. SAHP is the exception: its closed-loop rate is unstable across warm-starts and no constant κ transfers, so it is reported uncalibrated (κ=1, marked †).
- **Statistics**: roll-out seeds are averaged within each checkpoint; 95% CIs are t-based across checkpoints. Any batch failure or non-finite value aborts the run.

Prediction Quality

According to Table 1, SS2P2 is the strongest predictor on two of the three assets: it wins *every* prediction metric on SOL by a wide margin (0.27 vs. 0.53 overall NLL; 0.192 vs. 0.184 accuracy), wins overall NLL and accuracy outright on ETH (-0.87 vs. NHP's -0.81; 0.345 vs. 0.341), and ties NHP on BTC marks while conceding overall likelihood. It achieves this while remaining verifiably calibratable in closed loop (Table 3); no baseline combines both.

The remaining gap is in timing, and we can locate it precisely. A per-gap decomposition of timeNLL $= -\log \lambda(t_i) + \Delta u_i$ (event term + compensator mass) on BTC, the asset where NHP leads, shows that burst sharpness is *not* the issue: the state-space family's event terms beat NHP's (-4.95 SS2P2 vs. -4.23), and intensity-ceiling saturation is comparable across models ($\sim$25%). The entire gap is compensator mass in *quiet gaps*: mean per-gap rescaled mass is 1.88 (SS2P2) and 2.26 (S2P2) against NHP's near-ideal 1.03 (target 1), with the excess accruing in gaps beyond 0.1 s, which are rare at 38 ev/s but expensive (SS2P2 integrates 10.0 units of hazard across a 0.5–2 s gap where NHP integrates 6.0; S2P2 16.4). Figure 3 shows why: NHP's per-dimension decay toward a *learned steady state* tracks the falling empirical hazard of real gaps, while the state-space

model	BTC		ETH		SOL	
	overall↓	ACC↑	overall↓	ACC↑	overall↓	ACC↑
NHP	**-1.07±0.01**	**0.449±0.003**	-0.81±0.03	0.341±0.006	0.53±0.01	0.184±0.004
LSTM	-0.44±0.40	0.411±0.015	0.15±0.04	0.288±0.003	1.34±0.25	0.141±0.005
SAHP †	-0.93±0.01	0.412±0.001	-0.60±0.01	0.297±0.004	0.67±0.02	0.155±0.003
PCT-LSTM	-0.60±0.08	0.313±0.010	-0.34±0.10	0.248±0.007	0.89±0.03	0.105±0.004
S2P2	-0.39±0.25	0.439±0.002	-0.38±0.22	0.331±0.008	0.67±0.49	0.182±0.002
SS2P2	-0.87±0.12	0.448±0.001	**-0.87±0.08**	**0.345±0.003**	**0.27±0.05**	**0.192±0.001**

Table 1: Prediction per genuine test event (mean ± 95% CI over three training seeds; streaming lossless evaluator; overall = time + mark NLL). SS2P2 wins both columns on ETH and SOL and ties NHP on BTC accuracy (0.447 vs. 0.449). Not shown: S2P2's expected-time MAE is pathological on every asset (1.5, 10.2, 46.2 s vs. 0.03–0.05 s for every bounded or recurrent baseline), the intensity-collapse tail of the unbounded head.

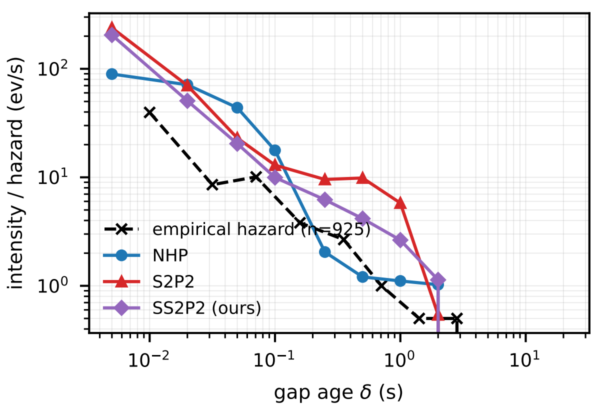

Figure 3: Mean model intensity at gap age δ, among BTC test gaps that survive past δ, vs. the empirical hazard of the real gap distribution (log-log; survivor counts shrink toward the right). NHP tracks the falling hazard; the state-space family stays hot through quiet gaps, where the timeNLL gap of Table 1 accrues.

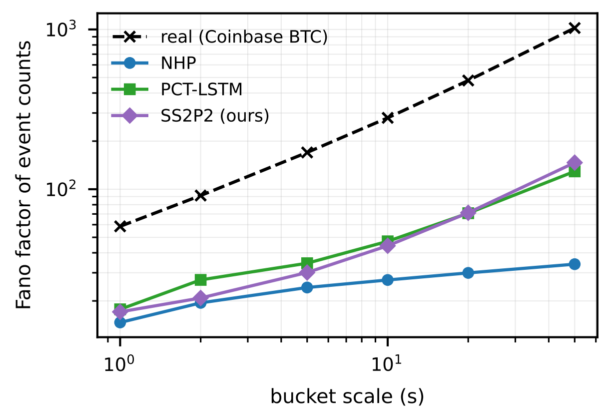

Figure 4: Fano factor of simulated event counts vs. bucket scale (Coinbase BTC; mean over calibrated checkpoints and roll-out seeds). Real order flow is super-Poissonian with Fano *rising* in scale; all models under-disperse at this context length, SS2P2 and PCT-LSTM least.

family, whose quiet-state behaviour is an emergent readout of decaying latents with no dedicated resting-rate parameter, stays hot through quiet gaps. This cost is the deliberate price of stability; what it buys is the subject of the next two subsections.

Simulation Quality

Table 2 and Figure 6 score calibrated closed-loop roll-outs on the stylized facts of real order flow. Every calibrated model lands its rate (rel-err 0.06–0.42); the discriminating columns are structural, and there SS2P2 is the only model that pairs competitive simulation with top-tier prediction: it is at or near the front on every structural column of every asset, and it leads clustering on SOL. Among the baselines, PCT-LSTM leads the Fano column but sits mid-table in prediction; NHP, the best BTC predictor, is mid-pack in simulation on every asset; and the LSTM pins Fano rel-err at ≈0.99 throughout, because it simulates a near-Poisson stream, the trivially-stable end of the trade-off. Prediction and simulation quality are distinct axes, and no unbounded model wins both.

Figure 4 shows the level-and-rise view on BTC: real Fano rises from ∼60 at 1 s to ∼1000 at 50 s; every model under-disperses at this context length, SS2P2 and PCT-LSTM least. The one surviving S2P2 arm on ETH calibrates in rate but over-disperses wildly (Fano rel-err 6.0): rate calibration cannot repair a near-critical checkpoint's structure.

Calibratability

Post-hoc rate calibration (Methods) applies to *every* decoder, because a constant κ rescales sim-time intensity without touching type probabilities, so we calibrate every model and treat the *outcome* as an experimental result. The protocol is deliberately falsifiable: probes must bracket and converge to the validation target, a full-scale roll-out must verify it, and every failure is reproducible (probe seeds are deterministic; each retried failure reproduced its bisection bracket bit-identically).

Table 3 is, to our knowledge, the first cross-asset quantification of closed-loop calibratability for neural TPPs: the bounded SS2P2 head verifies on *all nine* checkpoints, while

model	BTC			ETH			SOL		
	rate_re	Fano_re	clus_re	rate_re	Fano_re	clus_re	rate_re	Fano_re	clus_re
NHP	0.09±0.08	0.86±0.07	**0.27±0.04**	0.37±0.20	0.87±0.11	**0.52±0.23**	**0.19±0.03**	0.75±0.17	0.72±0.11
LSTM	0.10±0.09	0.99±0.00	0.50±0.41	0.40±0.03	0.99±0.00	0.77±0.29	0.23±0.04	0.99±0.00	0.97±0.03
SAHP †	0.71±0.38	0.77±0.19	0.93±2.49	0.79±0.12	0.82±0.23	0.34±0.73	0.68±0.15	0.92±0.08	0.65±0.25
PCT-LSTM	**0.06±0.06**	**0.79±0.12**	0.36±0.30	0.42±0.26	**0.76±0.17**	0.55±0.50	0.22±0.25	**0.46±0.34**	0.72±1.35
S2P2	0.32±1.51	1.25±12.45	0.29±0.14	**0.27**	6.05	0.60	–	–	–
SS2P2	0.25±0.43	0.81±0.37	0.29±0.17	0.42±0.17	0.79±0.05	0.81±0.11	0.30±0.17	0.60±0.60	**0.68±0.62**

Table 2: Closed-loop stylized facts (rel-err vs. real; roll-out seeds averaged per checkpoint, mean ± 95% CI over checkpoints). Checkpoints that failed calibration verification are excluded (BTC: S2P2-s1; ETH: S2P2-s1/s2; SOL: all three S2P2 seeds, shown "–"). † = SAHP uncalibrated by protocol; its free-running rate collapses to 6–10 ev/s against 24–48 real.

model	BTC 38 ev/s	ETH 48 ev/s	SOL 24 ev/s
NHP	3/3	3/3	3/3
LSTM	3/3	3/3	3/3
SAHP †	*model-level divergence: reported uncalibrated*		
PCT-LSTM	3/3	3/3	3/3
S2P2	**2/3**	**1/3**	**0/3**
SS2P2	3/3	3/3	3/3

Table 3: Checkpoints passing calibration + full-scale verification, out of three training seeds per asset (bold marks failures). The unbounded S2P2 head fails on 7 of 9 checkpoints, including *all three* on SOL where no usable simulator remains, while the bounded SS2P2 head verifies on all nine. SAHP diverges at the model level on every asset.

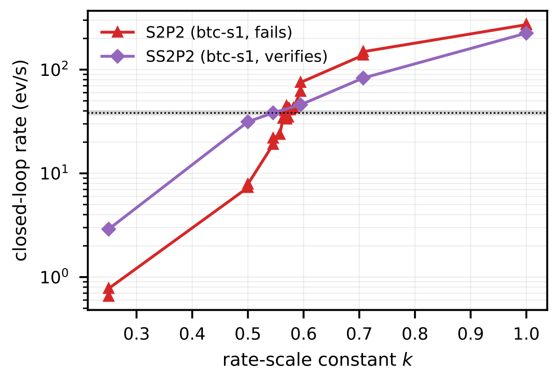

Figure 5: Calibration probe ladders on BTC (target band shaded): the failing S2P2 checkpoint's closed-loop rate is near-discontinuous in the rescaling constant κ; the verified SS2P2 checkpoint crosses the target smoothly.

the unbounded S2P2 head fails on 7 of 9, and on SOL the model class is excluded outright. Figure 5 shows the failure signature: the closed-loop rate of a failing S2P2 checkpoint is near-discontinuous in κ: a 10^{-3} change flips the realized rate across the entire tolerance band (e.g. $36 \rightarrow 48.5$ ev/s), and seed-to-seed variance at fixed κ reaches $\sim$30%. This is the behaviour of a system parked near criticality, where no constant time-rescaling exists. The SS2P2 response crosses the same target smoothly. (An earlier training run of one BTC seed produced a checkpoint whose single failure was qualitatively different: a per-lane diagnostic traced it to a *metastable* burst-locked regime at $\sim 2\times$ the target rate that individual roll-out lanes enter stochastically, bounded and benign next to S2P2's criticality; retraining escaped that basin.) This confirms the stability certificate empirically: the same unbounded softplus head whose intensity-collapse states produce S2P2's pathological expected-time tail (Table 1) also places checkpoints near criticality in closed loop, and both problems disappear under the softmin bound.

Efficiency Analysis

The state-space backbone is also computationally advantageous once its per-event sequential loop is replaced by an exact associative prefix scan (Hillis–Steele over the ZOH recurrence; identical states and gradients to 2×10^{-4}). Table 4 reports wall-clock measurements on one GPU with the benchmark configuration and ETH checkpoints: one training step (fwd+bwd, batch 64×1024 events) and closed-loop simulation throughput (simulated events per wall-second, 8×60 s carried-state roll-outs at native rates; SAHP re-encodes a sliding window). The scan collapses the S2P2-family training step from 3.5 s to $\sim$0.06 s: attention-class speed, $60\times$ faster than the loop and $73\times$ faster than the CT-LSTM, which turns a multi-hour training run into minutes. For sampling, SS2P2's exact ceiling makes Ogata thinning available with a *constant* dominating rate (no per-step bound estimation; mean acceptance 0.54 on ETH): exactness costs $\sim 2.5\times$ throughput against the approximate inversion sampler shared by all models, and both run far above real time (353 vs. 48 events/s on the fastest asset).

Discussion and Conclusion

We held six neural MTPPs to both sides of the prediction–simulation tension on three Coinbase assets at production event rates (24–48 events/s), under a protocol designed so that failures are visible and reported: lossless streaming prediction, multi-seed checkpoint-level confidence intervals, and verified rate calibration whose failures count as results. The SS2P2 heads (a softmin-bounded rate with a closed-form thinning ceiling and an exactly-zero floor, times a rate-neutral simplex mark head on an unchanged state-

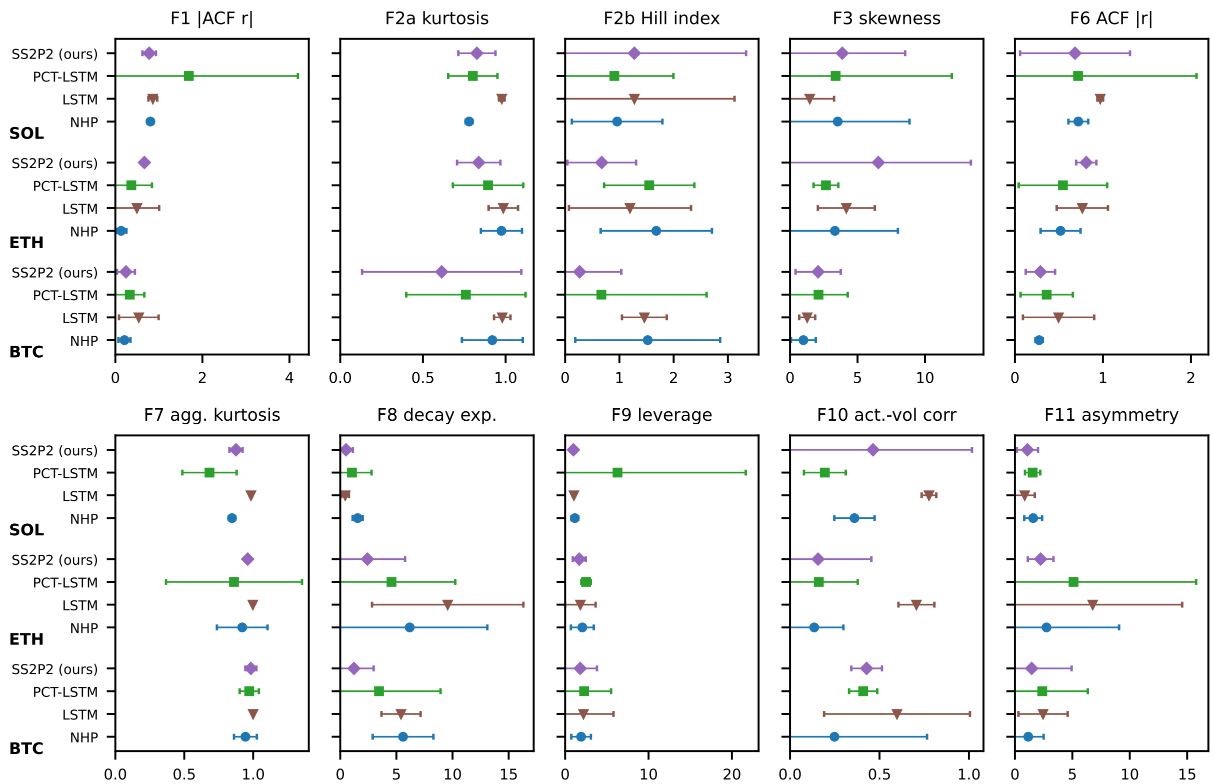

Figure 6: All ten stylized facts across assets: relative error vs. real (whiskers = 95% CI over checkpoints; roll-out seeds averaged within checkpoint). S2P2 and SAHP omitted where no calibrated checkpoints exist.

	train step (s)	vs. seq. loop	sim. ev/s
NHP	4.35	–	1253
LSTM	0.019	–	2083
SAHP	0.059	–	583
PCT-LSTM	0.79	–	1919
S2P2	0.055	64×	1163
SS2P2	0.059	60×	879 / 353†

Table 4: Efficiency on one GPU (ETH checkpoints, benchmark configuration). Training: one fwd+bwd on 64×1024 events (scan for the S2P2 family; speedup vs. their sequential loop). Simulation: events per wall-second, 8×60 s roll-outs. †inversion / exact-ceiling Ogata thinning.

space backbone) win every prediction metric on SOL, lead ETH, tie the strongest baseline on BTC marks, and remain verifiably calibratable with competitive simulated structure throughout. A per-gap hazard decomposition shows that the remaining prediction gap comes from quiet periods, the expected cost of bounding the rate. The sharpest finding is that *calibratability is a model property*: the unbounded parent head fails verified rate calibration on 7 of 9 checkpoints, including all three on SOL, with a critical-point signature, while the bounded head verifies on all nine. This confirms the stability certificate empirically. Stateful training and carried-state roll-out contribute temporal structure that calibration alone cannot. Because the mark head is rate-neutral, conditioning it on exogenous state such as an agent's resting quotes cannot destabilize the rate, so SS2P2 can serve as an environment model for market-making world models; that action-conditioned extension is future work. The same factorized intensity also supports the importance-sampling probabilistic queries of Chang, Boyd, and Smyth (2024), queries of the form "event A before event B, given the history," which map directly onto the quantities a market maker needs from a world model: fill probability (an aggressing order reaches our resting level before the quote moves away) and adverse-selection risk (the probability that flow continues against us conditional on a fill), estimated consistently from one likelihood-trained model rather than by separate ad-hoc regressions. All tables and figures are regenerated end-to-end by the released collection and display scripts.

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